The Law of Graphic Equalisation — Mind and Self Grounding

Frames, the Run and the Record, Free Will, Personal Identity, and Death

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Abstract

This companion to Graphic Equalisation develops the mind-and-self groundings of the change-substrate. Its central move is that a frame is real graph data that *runs*, and that everything here follows from the run and its record. A rock is a universe, so anything a rock does without is not compulsory for a universe. Recording spatialises a succession into a static held axis, giving the run/record axis whose poles are Conway (pure running, computationally irreducible) and a movie (pure record), with real frames the machinima between; the past can be a record but the future is uncomputed, so it is open as *unrun*, not as undetermined. Free will is then the in-frame run — a fully recorded, zero-randomness story demonstrates it, its characters free in frame while the reader foreknows — so freedom needs no indeterminacy. Imagination is real: there is no non-actual frame, and the only difference between the imagined and the perceived is frame-relative grounding. Death is the self's rules ceasing to be re-runnable in the same coherent self; personal identity is internal coherence (with an instance a supergraph node), which places gradual replacement, teleporter, backup, and fission on one axis; and death is relative to a chosen coherence-boundary, with many real deaths and none absolute. The claims are information-level and structural: the paper mechanises the shapes and offers no decision procedure, no operational test of when coherence is lost, and no theory of consciousness or qualia.

1 Scope

The method is to treat philosophy of mind as applied physics: once a frame is graph data that runs, free will, identity, and death are read off the run and its record, not argued as separate doctrines. This paper is a grounding, not a psychology and not a metaphysics asserted from within. It inherits Graphic Equalisation's position that the claims are about information — distinction, frame, admissibility, realisation, and now the *run* of a frame and its *record* — of which mind is one embodiment. The strong claims are structural: that free will is the in-frame run and needs no indeterminacy; that imagination is real and differs from perception only in grounding; that personal identity is internal coherence, distinct from the supergraph node that instances it; and that death is the loss of re-runnability in the same coherent self, relative to a chosen coherence-boundary. The weak part is anything operational: no decision procedure for a

given choice, no test for when re-runnability or coherence has in fact been lost, no threshold for how gradual a change stays "the same" self, and no account of the subjective, felt, or phenomenal character of any of it — consciousness and qualia are out of scope. Each grounding is a *promoted* constraint exposed to falsification, in the manner of Graphic Equalisation's collapse bridge, not a result asserted from within.

2 The minimal universe

A *rock*, in the sense used throughout, is an inert frame: a graph whose rules are present but not firing, so it has no succession and no time of its own. Things happen to it from its supergraph, but in its own frame it merely is (the physics companion develops this as why a rock has no time). Everything here is a graph, so a rock and a universe are one kind of thing, differing in scale, not in category (Graphic Equalisation's self-similarity: the runner is graph-that-runs at every scale). A rock *is* a universe — the minimal one — and that makes it an existence proof about universes in general.

Proposition 2.1 (Compulsory versus contingent)**.** *Anything not compulsory for a rock is not compulsory for a universe. The rock does without time (it is inert, no succession), without cycles or closure, without inference or observation, without history or records, without self-originated change, and without a running arrow — so none of these is compulsory for a universe. What even the rock cannot do without is the compulsory core: structure (it* is *nodes and relations), rules (graph data, present though inert), being a subgraph of a supergraph (which applies changes to it from above), and an admissibility grammar (the rules that would gate its self-change, unfired). That core is the whole of what a universe must have.*

This is the general form of "time is optional," and it licenses the contingent-not-necessary verdicts of the physics companion. Wherever a shape there was called contingent rule-content rather than necessity — the number of spatial dimensions, the gauge group, the dimensionless constants, the arrow, the value of c — the rock is the reason: the minimal universe does without it, so it cannot be compulsory. Physics ordinarily takes time, causality, and the second law's arrow as features a universe must have; the rock shows they are features an *active* universe has. And the block universe is not prohibited either: an MRI of a rock is a block-universe progression — the succession held as one static laid-out frame (Graphic Equalisation's history-as-held-frame). What Axiom −1 withholds is only a *completed* block with a fixed future already in it; a block of the succession *so far* is an ordinary held record, open at the frontier, not forbidden.

Honest boundary: the argument is as strong as the self-similarity it rests on (rock and universe as one kind, differing by scale). It does not claim a universe *is* inert like a rock — an active universe has time, change, an arrow. It claims only that a universe *need not* have what the rock does without: those features are contingent on activity, not compulsory to being a universe at all.

3 Running and recording: Conway, a movie, and the machinima between

Recording spatialises time. A succession held as a record — an MRI of a rock, a film strip, a worldline drawn on paper — lays the temporal order out as a static structure, a +-like axis (the block's extra dimension). This reads the metric signature: the − axis is the one being *lived*, the frame's own succession run by its cycles, open at the frontier; the + axes are *held* — spatial positions, and any recorded time. The sign tracks lived-versus-held, not a property intrinsic to the axis; the block feels dead because a recorded succession is not run.

The extremes of the run-versus-record axis are Conway and a movie. Conway's Game of Life is pure running: a rule and a seed, everything computed live, nothing recorded, and no

way to know the outcome but to run it (computational irreducibility). A movie is pure record: every frame laid out, played back, nothing computed — the block universe, known by reading rather than running. Between them is scripted machinima — a running engine following a predetermined script, part computed live, part laid down in advance — and that is where real frames sit: rules fire (Conway-like) while history is held (movie-like). It is the same axis as generation-versus-record: store the generator and run it (Conway, small but irreducible) or store the frames and read them (a movie, large but immediate), with the world's own compressibility deciding which is cheaper.

Proposition 3.1 (The past can be a movie; the future is Conway)**.** *A block is only ever of the past. What has been run can be recorded and laid out — a movie of the succession so far, readable without re-running. What has not been run cannot be recorded, because there is nothing yet to record: the future is determined by the rules but* uncomputed*, Conway-style, so it is not laid out and cannot be read ahead — you must run it. The open frontier of Axiom* -1 *is therefore not undetermined-ness but unrun-ness: the future is open because it is uncomputed, not because it is unfixed, and you cannot record what has not run.*

Honest boundary: this applies Graphic Equalisation's generation-versus-record axis to whole universes, with Conway and film as its two poles (Conway is itself a seed in the substrate). It adds no formal machinery — it identifies the lived/recorded and generation/record axes as one, and locates the open future as the unrun edge of the run, not as indeterminacy.

4 Free will and the three frames

Corollary 4.1 (Free will is the in-frame run; a story demonstrates it)**.** *Free will is a frame running its own decision. In a frame's own run the decision is live and unrun-to-it — the frame must run its deliberation to reach the outcome — and that running* is *the choosing: the outcome is fixed by the rules yet knowable only by running, so the frame experiences computing it as choosing it. This needs no indeterminacy, and a story is the perfect demonstration. A book or a movie is a fully recorded succession — every word or frame fixed, zero randomness, readable ahead by the reader — and yet its characters have complete free will: they deliberate, choose, and are responsible, in frame. Both hold at once because they are two frames. The character's freedom is its in-frame run; the reader's foreknowledge is a superframe holding the finished record (a movie), and holding the record never reaches into the character's run to remove the freedom. So foreknowledge is not unfreedom — not even complete foreknowledge, the reader who knows the ending or the author who wrote it — because knowing the recorded run and being the live run are different frames. Randomness would only subtract: a random run is not more free, it is less the character's own. And responsibility survives for the reason the freedom does — the run is the character's own (self-change is the only change; the rules running are the character) — so a determined, recorded, even foreknown act is still that character's act.*

Honest boundary: this is a compatibilist stance — free will as the frame's own in-frame run, frame-relative and robust to a superframe's complete foreknowledge, with no appeal to indeterminacy. It claims the shape of the free/determined dissolution, not a decision procedure, and takes no position on which acts a given reader can in fact foreknow (that is the run/record split of Proposition 3.1).

Remark (Author, character, reader: unrun-ness is absolute). A story stacks three frames on one succession. The author *runs* it — writing is Conway, generated live, and the author must write to know the ending (authorship is computationally irreducible; the commonplace that "the characters took on lives of their own" is the author unable to read ahead their own creation). The character *lives* it — the in-frame run of Corollary 4.1. The reader *replays* it — the finished record, a movie. One succession, three frame-readings: the run/record axis is a fact about

frame, not about the system. And the author settles the objection to Proposition 3.1 that a high enough superframe might read the future ahead — even the creator gets no shortcut. The author is Conway to their own book while writing it; the movie exists only *after* the run, for everyone including the author. Foreknowledge is always post-run record-holding, never pre-run read-ahead. Unrun-ness is absolute: no frame, creator included, reads an unrun future; every frame either runs it or replays an already-run record.

5 Imagination, grounding, and reality

Remark (There is no non-actual frame; imagination is real)*.* Imagination is not the running of a non-actual frame — there is no such thing. Every frame that runs is real graph data really running, so actuality is not a property some frames have and others lack; every frame is actual. An imagined dragon, a planned move, a character in a book is a real frame, really run (real computation), not a pretend one. The only difference between an imagined frame and a perceived one is frame-relative grounding: what the frame is anchored to, relative to another frame. A perception is grounded in the observation frame (sensory-anchored); an imagination is grounded internally, in generation or in a held or communicated record. Both are real; they differ in what they are grounded in, not in whether they are. So "is it real?" is malformed as a question about actuality — everything that runs is — and the well-formed question is "what is it grounded in?": observation, memory, generation, or a communicated record. A hallucination is then a real frame mistagged as observation-grounded (a grounding error, not a false frame); a shared fiction is a real frame grounded in a communicated record (the book of Remark 4); a plan is a real frame grounded in projection. This is why imagination does work — planning, design, and counterfactual reasoning are real frames really run, not pretence — and why fiction moves: the character's frame runs, really, in the reader. Death, in the same terms, is not a frame going non-actual (nothing does) but a run that stops, leaving frames grounded only in a record and re-runnable by others; reading and remembering are that re-running — a record-grounded frame run again into a live grounding.

Honest boundary: actuality is not a frame-property (every running frame is actual); the real distinction is frame-relative grounding, and this remark does not claim every frame is equally grounded in observation — the dragon is not — only that every frame is equally real, differing in what it is grounded in. It claims the shape (reality is running; the distinction is grounding), not a metric of grounding.

Corollary 5.1 (Grounding has no bedrock; observation is supergraph-contact)**.** *Grounding is frame-relative all the way down: a frame grounded in a frame grounded in a frame, with no absolute, self-grounding frame beneath them all. There is no bedrock "actual" layer — the ground is the coherent running structure, frame to frame, not a base. Yet one grounding is privileged, relationally and not absolutely:* observation *is a frame's contact with its supergraph, the boundary where the container deforms it with changes it does not author (Graphic Equalisation's dissolved pointer — you change yourself, the supergraph deforms you — with perception the deformation side). Observation is special because it is* yours*, your container-boundary, not because it is "the real." It follows that reality-testing is not actuality-checking — there is no non-actual frame to catch (Remark 5) — but* supergraph-pushback*: does the container deform you as if the frame were grounded there? You walk into the chair (pushback: observation-grounded) and not into the dragon (no pushback: imagination-grounded), both real frames, only one with the container pushing back. Sanity is then a grounding-tag problem, not a reality one: keep observation-grounded frames tagged as observation-grounded and imagination-grounded ones as imagination-grounded. A hallucination or delusion is a grounding-tag error — a real, internally-grounded frame mistagged as observation-grounded, so the self expects a pushback that will not come — not a belief in something non-actual, since nothing is. The position is*

thus anti-foundationalist (no absolute ground) without being flatly coherentist (observation is privileged, as your supergraph-contact).

Honest boundary: grounding is frame-relative with no bedrock, and observation is privileged only relationally, as the frame's supergraph-contact; reality-testing is container-pushback, not actuality-checking. This claims the shape (ground is supergraph-contact; testing is pushback), not a perception model, and it does not deny that the container pushes back hard and reliably — only that its pushing is what "grounded in observation" means, rather than a separate property of being real.

6 Death and personal identity

Remark (Death is when the self's rules cease to be re-runnable in the same coherent self)*.* Death is, critically, the rules within the self ceasing to be re-runnable in the same coherent self entity. Not every stop is death: a pause — sleep, anaesthesia, a recoverable coma — is a stop whose rules remain re-runnable in the same coherent self, and the self resumes as itself. Death is the stop that does not resume: the coherence that made these rules one self is gone, so they cannot be re-run as that self, even if the graph-data they were persists scattered as a record. This is the identity criterion — the self persists exactly while its rules stay re-runnable as the same coherent entity — and it is why a pause is survivable and death is not; the self-graph goes inert for good, a record where it was a run (the rock of Section 2, now grounded only in what remains of it).

What ends is only the self's own run. The supergraph runs on, and two things the self did persist in it. Its changes persist: rule-actions are pointers, work done once, so every write the agent made to the supergraph is the supergraph's now, permanent, and death does not undo it. And its copies persist: every frame the agent communicated, taught, wrote, or built runs on in other frames, grounded there in the record it left (the reader re-running the author of Remark 4). But those copies are partial fragments, not a full coherence-snapshot: they run as fragments in other coherent selves, not as this coherent self re-run, so their persistence is not the self's continuation (a full snapshot is the different case of Corollary 6.1, where the self does re-run, forked to a new node). The self can no longer self-change, and self-change was the only change it had; the changes it already made and the copies already running elsewhere are no longer its to stop. They keep running without it.

Honest boundary: this is the run/record, pointer, and copy machinery applied to a self, and it describes what death *is* in the substrate — the self's rules ceasing to be re-runnable as the same coherent self, while its writes and its copies persist — not any continuation of the self's own run. There is no such continuation: nothing re-runs as the same coherent self, and what persists (consequences and copies) is not that self running. The identity criterion is offered as the discriminator between a pause and death, not as an operational test of when re-runnability has in fact been lost; and this claims the shape, not an account of the subjective experience of dying.

Corollary 6.1 (Identity is internal coherence; instances are supergraph nodes)**.** *Two things must be kept apart: the self's* internal coherence *— its rules and state — and its* supergraph node *— the instance of it running at a position in the container. The self is the internal coherence; a node is one instance of it. This resolves the standard puzzles. Gradual replacement (cell turnover, the Ship of Theseus) is survival: internal coherence is continuous and there is one node throughout — the self is the coherent running, not the substrate. A backup restored is still the same internal self — a full coherence-snapshot re-run reinstates that exact coherence — but now at a* second *supergraph node, and from the restore point the two diverge, each accruing its own changes under its own grounding. That is a fission: one internal self, two supergraph nodes, diverging, both the self at the fork and neither uniquely the node after it. A teleporter is the same with one branch pruned — the full snapshot reinstates the internal self at a new node, the original node*

stops — so the internal self survives, relocated. What matters about a record is whether it is full *or* partial*: a full coherence-snapshot re-runs into the same internal self (forked to a new node), while a partial record — a corpse, a memory, a written portrayal — cannot reconstitute the coherence, so re-running it starts a new coherent self, not the original. The internal self therefore persists exactly as long as a re-runnable full coherence-snapshot exists somewhere; the death of a node is one instance stopping, and the death of the self is that re-runnability being gone everywhere.*

Remark (Death is relative to the coherency of the self)*.* It follows that death is relative to the coherency of the self. The criterion applies within a coherent self, but which coherence counts as "the same self" is a frame-relative choice, so death is indexed to that choice, not absolute. Drawn narrowly — this instant's exact coherent state — the self dies each moment, replaced by the next; drawn broadly — the coherent person tolerant of change across a life — it dies once, at the irreversible break. A deep transformation ("I am not the person I was") is a death relative to the narrow coherence and a survival relative to the broad one, both true at once. There is no absolute self, so there is no absolute death — only a death per coherence-boundary. This makes death relative, not unreal: each boundary's death is real for that boundary — the broad life-self's coherence does break irreversibly, and that death is as real as any — so relativity means many real deaths at many boundaries, never none.

Honest boundary: identity is the self's internal coherence, re-runnable from any full snapshot, while an instance is a supergraph node (a shape, not a sharp threshold for how gradual a change counts as unbroken, nor an account of what physically counts as a full snapshot); and death is relative to a chosen coherence-boundary — and to the node/self distinction — with many real deaths, none absolute, which relativises *when* death occurs and to which self, without denying *that* it occurs at each boundary.

7 The one assumption removed

The simplicity of every dissolution above comes from a single refusal: that humans — or any frame — are privileged. It is the most pervasive assumption in these problems and the least noticed, and each hard problem was made hard by smuggling it in. The measurement problem privileges a human (or conscious) observer as the special coupling that collapses; the flow-versus-block dispute privileges human-felt passage as the real time; the free-will problem privileges human choice as needing a metaphysical freedom no other system has; the identity and death problems privilege the human self as a persisting soul rather than a coherent run; the reality-versus-fiction and foundation problems privilege human perception as access to "the real." Remove the privilege — treat a human frame by the same frame-relative rules as a rock, a story character, or a universe — and each collapses into the general machinery already built: any coupling measures, time is any active frame's cycle-count, free will is any frame's in-frame run, the self is internal coherence, and observation is one frame's supergraph-contact rather than a window on the real. The physics companion's dissolutions (the special observer, the initial state) are the same refusal in the other domain.

The assumption persists because it is uncomfortable to drop. Unprivileged means not the centre, mortal as the loss of a coherence, free only as any frame is free, and never in contact with "the real" except as one's own container-boundary. The discomfort is why the assumption is both pervasive and invisible: it is motivated, held rather than observed. So the framework is simple not because its answers are clever but because the difficulty was manufactured by the privilege, and removing it removes the difficulty. This is Graphic Equalisation's Axiom -1 applied to observers: no frame is the god's-eye, a human frame makes no privileged difference, and "you are here" is a location, not a throne.

Honest boundary: this is a methodological claim about *why* the dissolutions are simple, not

a further mechanism, and it is not the claim that human frames are nothing. Human frames are distinctive — rich aperture, language, self-models — but distinctive is not privileged: being one remarkable frame among many is not being the exception to the rules the rest of them run under. No-privilege removes the exemption, not the richness.

8 Positioning and prior art

The positions here are not new as positions; what is new is that each falls out of one substrate mechanism — the run, and its record — rather than being posited on its own. On free will this is a compatibilism, determinism and freedom coexisting in the tradition of Hume (1748) and Dennett (1984), but with a specific mechanism for the freedom: the in-frame run and its computational irreducibility (Wolfram, 2002), so that the future is open as *unrun*, not as undetermined, and a fully recorded, zero-randomness story still exhibits free will. On personal identity the continuity criterion is Lockean in spirit (Locke, 1690) and closest to Parfit (1984) — identity as psychological or functional continuity, fission that branches, and what matters being the continuity rather than a bare identity fact — with the addition of the split between internal coherence (the self) and supergraph node (the instance), which places teleporter, backup, and fission on one axis. On time the past-as-record / future-as-run asymmetry engages the block-universe debate (McTaggart, 1908; Putnam, 1967): a block of the past is an ordinary record, while a completed block with a fixed future is not, because the future is uncomputed. And the no-absolute-self reading of death — a death per coherence-boundary, the narrow self dying each moment — recovers a momentariness familiar from other traditions, here as a consequence of frame-relativity rather than a doctrine. The distinctive claim throughout is the derivation, not the destination.

9 Falsifiability commitments

- **Compulsory versus contingent.** Fails if some feature the rock demonstrably lacks (time, history, self-change, a running arrow) is nonetheless shown compulsory for any graph to exist — if the minimal universe (an inert graph) is incoherent as a graph. The argument is only as strong as the self-similarity it rests on (rock and universe one kind, differing by scale).
- **The block is always of the past.** Fails if a completed block with a fixed future is coherent under Axiom −1, or if a block of the succession-so-far (an MRI of a rock) is forbidden. The claim predicts records only ever trail the run, never lead it.
- **The future is unrun, not unfixed.** Fails if a determined future can be read ahead without running it — a shortcut past computational irreducibility for a genuinely irreducible rule — or if a block can contain a not-yet-run future.
- **Free will is the in-frame run, not indeterminacy.** Fails if a fully recorded, zero-randomness system (a story) cannot exhibit the form of free will — deliberation, choice, responsibility — in its characters' frames, or if a superframe's complete foreknowledge demonstrably removes a subframe's in-frame freedom. The claim predicts foreknowledge and freedom coexist across frames.
- **Actuality is running, not a frame-property.** Fails if some running frame is demonstrably non-actual (real graph data that runs yet is not real), or if the imagined/perceived distinction requires an actuality property rather than frame-relative grounding. The claim predicts imagination does real computational work (planning, design) as real runs, not pretence.
- **Grounding has no bedrock; reality-testing is supergraph-pushback.** Fails if there is an absolute, self-grounding frame beneath all others, or if reality-testing can distinguish a

real frame from a non-actual one by anything other than whether the container (supergraph) pushes back. The claim predicts hallucination and delusion are grounding-tag errors, not false beliefs about actuality.

- **Death is loss of re-runnability in the same coherent self.** Fails if a pause that preserves re-runnability as the same coherent self (sleep, a recoverable coma) counts as death, or a stop that loses it counts as survival — the criterion is re-runnability in the same coherent entity, not the mere stopping. Also fails if a self's writes to the supergraph vanish when its run stops (work-done-once violated), or if frames it copied into other frames cease when it does. The claim predicts consequences and copies persist, that copies are not the self resuming, and that nothing re-runs as the same coherent self.

- **Identity is internal coherence; death is relative to the coherence-boundary.** Fails if the self is the substrate parts rather than the internal coherence (gradual replacement must survive; a full coherence-snapshot restored must be the same internal self forked to a second node, diverging; a partial record re-run must be a new self; a teleporter must relocate the same internal self). And fails if there is a single absolute self whose end is the one death independent of any coherence-choice, or if no coherence-boundary's death is real — the claim predicts many real deaths at many boundaries and levels (node versus self), none absolute.

- **No frame is privileged.** Fails if any dissolution here secretly requires a human frame to be privileged — a special observer, a special time, a special freedom, a persisting soul, a window on the real — rather than following from the frame-relative machinery applied uniformly to every frame. The claim is that removing the privilege, not any added mechanism, is what dissolves each problem.

- **Scope.** The paper claims shapes only; it is falsified as over-claiming if any section is read as a decision procedure, an operational test of when re-runnability or coherence has been lost, a threshold for how gradual a change stays the same self, or a theory of consciousness or qualia — none is offered.

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